https://github.com/bardia2532/Advanced-Data-Mining-D7043E

Customer Personality Analysis

CRISP-DM for Advanced Data Mining (D7043E)

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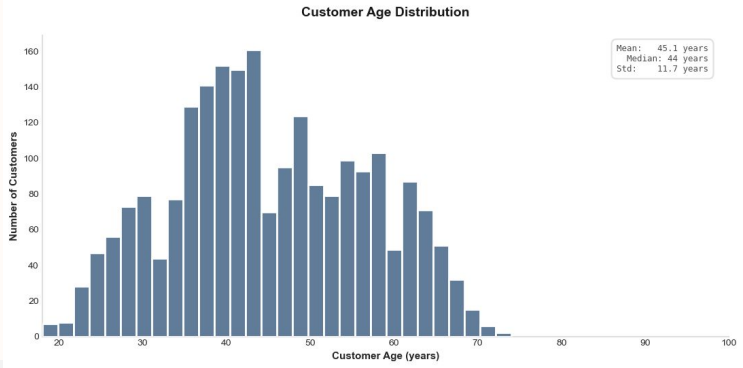
Task 1: Business, Data, and Preparation

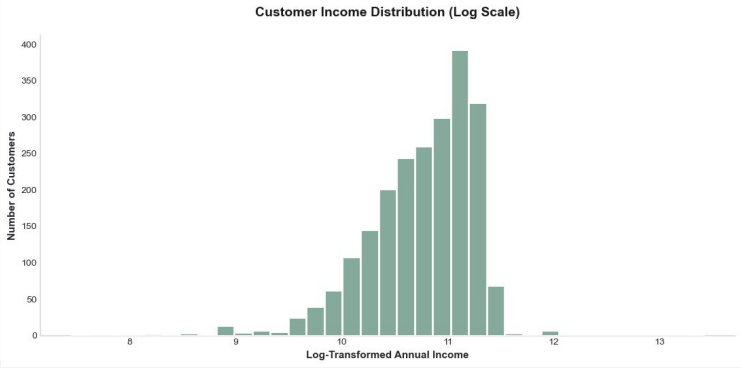
What we did:

- Defined two project goals: campaign response prediction and customer persona clustering.
- Explored and cleaned customer data (2,216 records, 27 features) to ensure quality.
- Engineered 51 new behavioral and demographic features for deeper modeling.
- Conducted extensive exploratory analysis with visual insights on spending, engagement, and campaign response.

Key features engineered (sample):

- `Total_Spent`, `Spending_Ratio`, `Avg_Spend_per_Purchase`
- `Family_Size`, `Income_per_Capita`, `Spend_per_Capita`
- Channel shares: `Web_Share`, `Catalog_Share`, `Store_Share`
- RFM: `Recency_Bin`, `Frequency_pm`, `RFM_Total`
- Flags: `Income_imputed`, `Outlier_flag`, `Zero_Purchases`





Goal A - Classification: Predict Campaign Response

Setup:

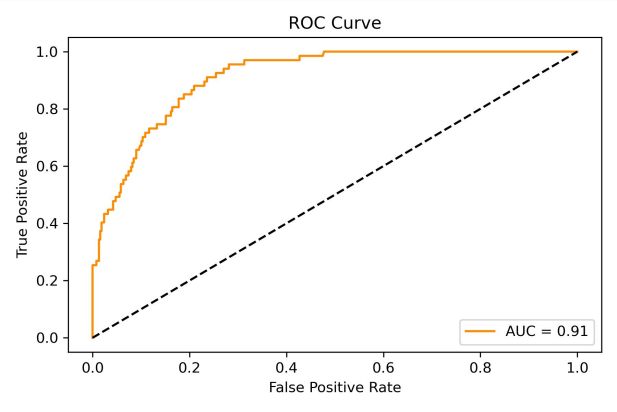
- Dataset: `customers_featured.csv` (features from Task 1).
- Target: `Response` (binary).
- Split: 80% train / 20% test, stratified.
- Preprocessing: median impute, scaling, one-hot for cat.

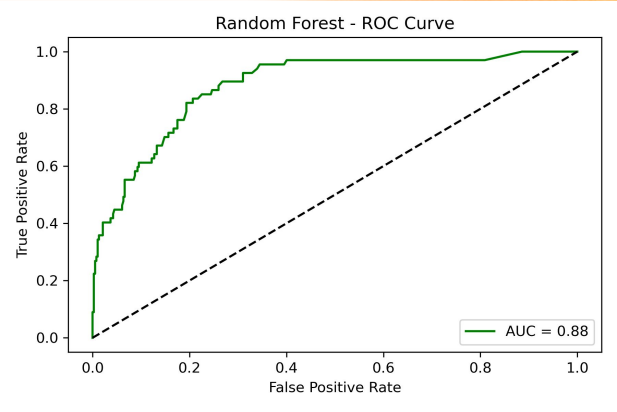
Models & key metrics:

- **Logistic Regression** - Accuracy **0.8919**, AUC **0.9120**.
- **Random Forest** - Accuracy **0.8919**, AUC **0.8815**.
- **Deep Logistic Regression** and **Deep Random Forest** - Attempted but failed due to class imbalance.

Business interpretation:

- The models identify which customers are most likely to respond to future campaigns.
- Both achieved around **89% accuracy and strong AUC**, indicating reliable prediction of responders.
- These results can help **prioritize customers for targeted marketing**, improving efficiency and return on campaigns.

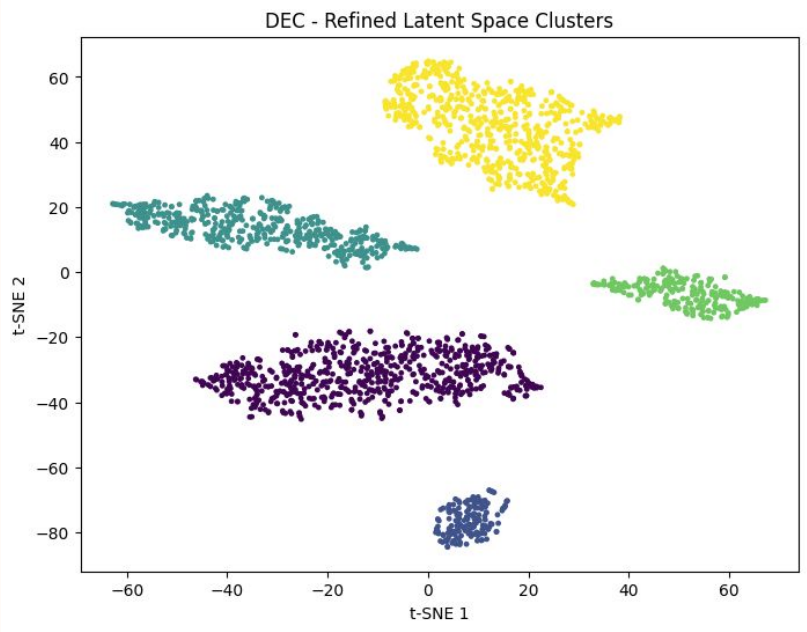




Goal B - Clustering: Personas via Deep Models

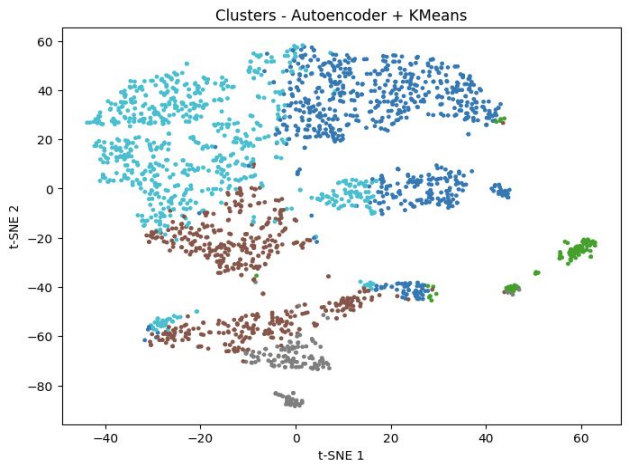
Methods used:

- **Autoencoder + K-Means** for baseline latent clustering.
- **Deep Embedded Clustering (DEC)** for refined, end-to-end clustering.



Quantitative comparison:

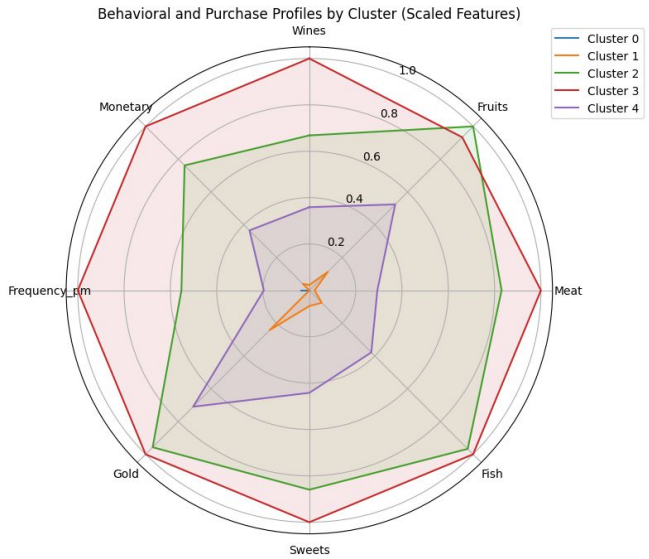
Metric	Autoencoder + KMeans	DEC
Silhouette	0.2408	**0.8845**
Calinski-Harabasz	740.51	**8164.64**
Davies-Bouldin	1.2473	**0.1788**

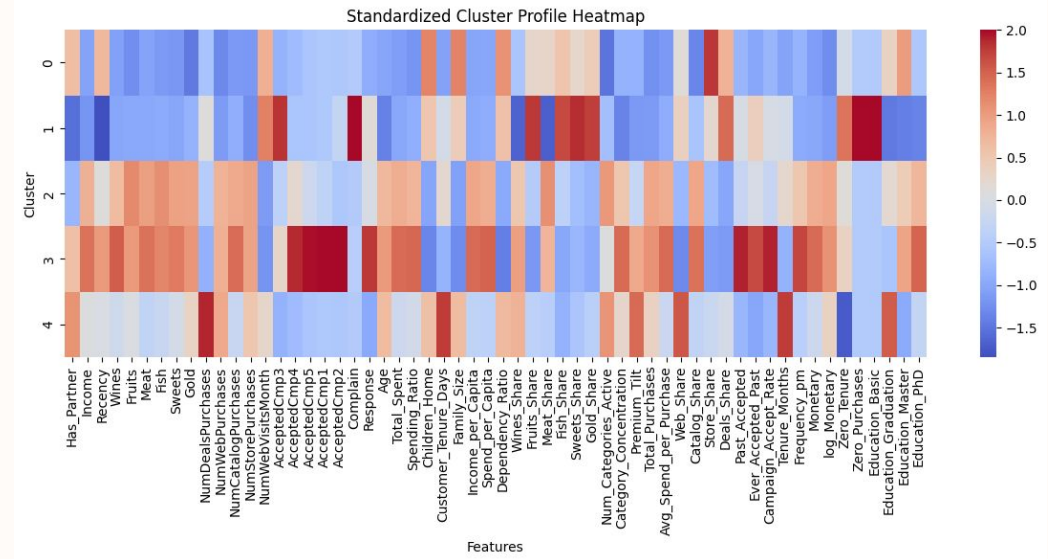


Cluster Profiles and Suggested Marketing Actions

Summary of top clusters:

- **Cluster 3 (Affluent, engaged):** High income, highest spending. Action: VIP offers.
- **Cluster 2 (Premium consumers):** High spend, selective offers. Action: exclusive catalogs.
- **Cluster 1 (Deal-seeking mid buyers):** Frequent purchases, deal-sensitive. Action: flash sales and coupons.
- **Cluster 0 (Moderate spenders):** Balanced preferences. Action: loyalty bundles.
- **Cluster 4 (Catalog-preferring):** Catalog + store focus. Action: improved catalog personalization.





Conclusions, Limitations, and Next Steps

Key takeaways:

- Task 1 supplied a robust, feature-rich dataset for Task 2 models.
- Classification models give strong predictive power (AUC > 0.88).
- DEC produced clean personas with high silhouette.
- Logistic regression helps business interpretation while DEC offers marketing segmentation power.

Limitations & cautions:

- Possible residual bias from income imputation.
- Clustering depends on chosen features and scaling.
- Models need out-of-sample validation before deployment.
- Deep Learning versions of the classification models need to be precisely tuned for better results.